



10953 - Building a Database of Models to Interpret JWST Nebular Type Ia Spectra

Cycle: 5, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

In cycles 1-4 JWST has awarded a over 90 hours to observing Type Ia supernovae (SNe), predominantly in the nebular phase. This epoch may be the key to finally identify the progenitors of these SNe, however theory is still struggling to keep up with this new window into these explosions. Before the launch of JWST most theoretical Ia spectra were only modeled in the optical, and even after the initial exciting results from JWST indicating that core asymmetries have been present in all five of the published examples to date, the vast majority of modeled nebular spectra available in literature have been performed in 1D.

The goal of this proposal is to perform a series of radiative transport simulations of Type Ia SNe to produce multi-D nebular emission spectra for optical through the mid-IR wavelengths. We will include a wide variety of SN Ia progenitor models from published literature creating the largest database of synthetic spectra to date. These models will be made publicly available to help the ongoing efforts to understand these explosions as they are observed with JWST.

OBSERVING DESCRIPTION